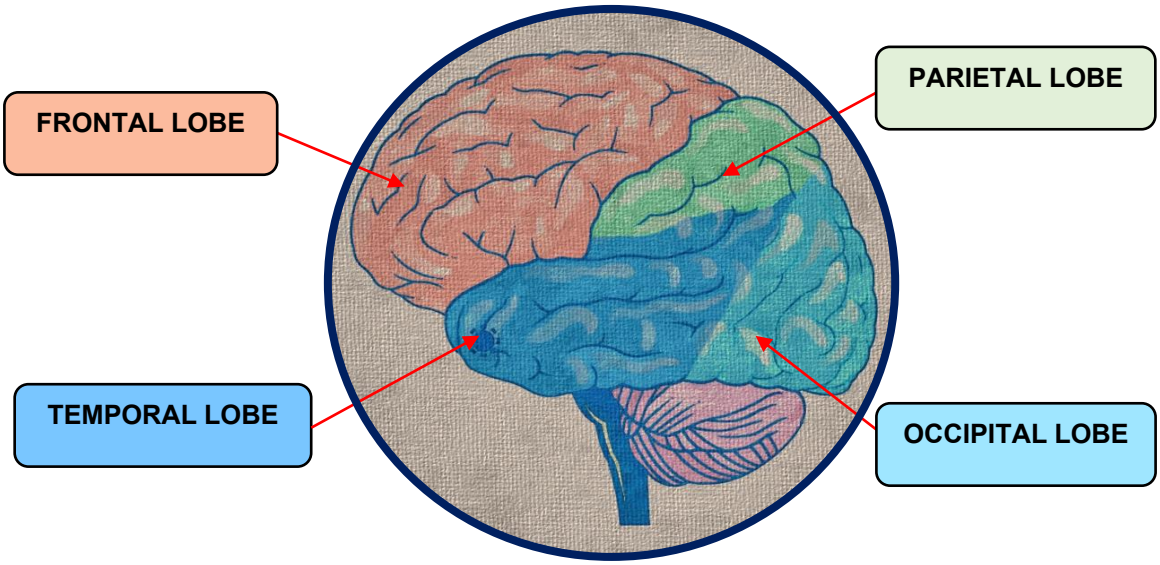
Section 2: Overview of Brain Function and Mutual Causality

The brain, which primarily consists of the left cerebral hemisphere and the right cerebral hemisphere, is a heterogeneous structure that is composed of neuroepithelial cells, blood vessels, and cerebrospinal fluid. The left cerebral hemisphere performs logical functions associated with deductive reasoning, pinpointing minute details, exercising caution, establishing order among randomness, and generating speech. The right cerebral hemisphere, which operates interdependently with the left cerebral hemisphere, performs cognitive functions associated with creativity, intuition, visual gestures, and imagination. This interdependence between cognitive functions of the right cerebral hemisphere and logical functions of the left cerebral hemisphere, which depends on the transmission of nerve signals across the corpus callosum, lays the groundwork for the learning process.

Candidate learning occurs in the following brain regions of the two (2) aforementioned hemispheres:

- Parietal Lobe: The right parietal lobe controls the ability to accept reason and to reproduce visual constructions. The left parietal lobe controls the ability to distinguish between right and left, to interpret objects (E.G. script in reading passages), to create objects (E.G. written scripts), and to establish mathematical relationships.
- Temporal Lobe: The right temporal lobe controls the ability to recognize auditory mechanisms, to channel emotional behavior, and to recall non-verbal material from memory. The left temporal lobe controls the ability to comprehend information from visual sources or verbal sources, to build awareness normally due to fear, to exercise caution, and to recall facts from memory.
- Frontal Lobe: The right frontal lobe controls the motor functions associated with cognition. The left frontal lobe controls the motor functions associated with language delivery.
- Occipital Lobe: The right occipital lobe and the left occipital lobe are responsible for generating vision in the proper color rendering on the left side and the right side, respectively.



The EvalzNow interface, Mind Tools, is a coarse mapping indicator for neuroscience experts to link human competencies of a given performance point to the major lobes of the cerebral cortex via relevant cognitive functions at the functional level. Mind Tools can supplement neuroscience studies.

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To create mathematical models that address the static/dynamic interdependency of the two (2) cerebral hemispheres, as well as changes in competency during the student learning process, a set of independent variables enveloping the full scope of interactions must be identified. An interaction, by definition, occurs when an independent variable has a different effect on the outcome depending on the values of another independent variable. Based on the functionality of the brain regions, the independent variables representing the interactions of the two (2) cerebral hemispheres and the student learning process have to be mutually causative and quantifiable. In other words, these independent variables must consider multi-directional cause-and-effect relationships, random or sequential, that can be measured directly through attainment – not necessarily regression or predictive analysis. This can be achieved by looking at the brain from a closed-loop functional perspective.

After the mathematical models are created, a unique learning profile can be generated per candidate to determine strengths and weaknesses, subject to interactions and competency changes. When weaknesses cause poor interactions and negative competency changes, candidates are normally experiencing either a learning difficulty/deficiency or a high competency/cognitive load. Users of EvalzNow should not confuse these weaknesses with unattainable challenges, especially if the candidate demonstrates feasible interactions and reasonable competency changes for attainable challenges.

Learning difficulties normally include:

- A.D.D./A.D.H.D.
- Asperger's Syndrome
- Dyscalculia
- Dysgraphia
- Dyslexia
- Impairments in Auditory Memory
- Impairments in Visual Memory
- Impairments in Working Memory
- Instabilities in Executive Function
- Lapse in Reasoning
- Loss in Auditory Processing
- Non-Verbal Communication

Mutual Causality

Mutual causality, which is derived from the theory of autopoiesis, refers to the recursive set of interaction chains that are self-produced within a closed-looped system. From a cognitive science perspective, the recursive set of interaction chains is directly linked to the independent variables that address the interdependency of the two (2) cerebral hemispheres, and the closed-loop system gives insight into the structural interactions between the two (2) cerebral hemispheres via the corpus callosum. Since the closed-loop system is self-produced, there are neither outside mechanisms nor external entities that act on the closed-loop system during operation. This means that the closed-loop system relies on internal mechanisms or internal entities to determine what interactions need to take place and when these interactions need to take place to obtain a desirable result during operation.

There must be at least two (2) independent variables to have an interaction in the closed-loop system. Moreover, the interaction among the independent variables must be based on the capacity of the closed-loop system to change, to adapt, or to modify its functional responses/connectivity to new information, expansion of the knowledge base, sensory stimulation, and dysfunction. Such capacity is driven by functional elasticity for temporary deviations and functional plasticity for permanent deviations. All users should always consider the capacity of the candidate within the following contexts:

- Abstract Reasoning
- Auditory Speech Discrimination
- Broca's Speech Pronunciation
- Kinesthetic Perception
- Kinesthetic Speech
- Lexical Memory
- Linear Predictive Speech
- Mechanical Reasoning
- Narrow Visual Span
- Non-Verbal/Artifactual Thinking
- Numerical Quantification
- Numerical Reasoning
- Object Recognition
- Primary Motor
- Short-Term Memory
- Long-Term Memory
- Spatial Reasoning
- Symbol Recognition
- Symbol Relations
- Symbolic Thinking

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